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Procedia Computer Science 228 (2023) 826–837

Procedia Computer Science

www.elsevier.com/locate/procedia

3rd International Conference on Machine Learning and Big Data Analytics for IoT Security and Privacy

Research on Traffic sign recognition based on CNN Deep Learning Network

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Abstract

With the continuous development of social economy, the road traffic system as one of the basis of national economy is also increasingly developed, which is followed by the traffic safety problems are increasingly prominent. Among them, driving violations such as not obeying traffic signs are the main cause of traffic accidents. In this research, Traffic Sign Recognition (TSR) has become a hot research topic. The research status and application field of traffic sign detection and recognition technology are introduced and analyzed in detail, and the improvement of traffic sign database is introduced. After that, the application of CNN in image processing and traffic sign recognition is introduced, and the existing problems in traffic sign recognition are analyzed. The CNN deep learning network used in this paper can conduct training while identifying targets, effectively improve the recognition accuracy of traffic signs under complex natural conditions and reduce the time consumption.

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Peer-review under responsibility of the scientific committee of the 3rd International Conference on Machine Learning and Big Data Analytics for IoT Security and Privacy

Keywords: Convolutional Neural Network, CNN, traffic sign recognition,deep learning

1. Introduction

The growing socio-economic development of our society has been accompanied by the development of the road traffic system, which is one of the foundations of the national economy, and with it, the growing importance of traffic safety issues. The most important cause of traffic accidents is driving behaviour such as disobeying traffic signs. The study of traffic sign recognition has become a major research hotspot. Traffic sign recognition is an integral part of intelligent driving systems and has an important place in the fields of assisted driving, driverless driving and traffic sign maintenance management. An effective TSR system can provide real and accurate real-time

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10.1016/j.procs.2023.11.102

traffic information back to drivers in a timely manner, prompting them to take corresponding measures or vehicle autonomous control to achieve risk avoidance, thus ensuring driving safety.

In more complex natural scenes, various external factors make it difficult to complete traffic sign recognition accurately and quickly, and the traditional methods of traffic sign recognition are already insufficient in terms of accuracy as well as speed to meet the current needs of driverless systems. Therefore, under the current trend of intelligent transportation development, there is a greater need for more scientific and advanced tools for traffic sign recognition. Traditional methods that rely on colour and shape features to match images are simple, convenient and fast, but the recognition rate and accuracy are difficult to achieve in practice. The CNN deep learning network used in this research could be trained while recognizing the target, which effectively improves the recognition accuracy and reduces time consumption, and can improve the correct recognition rate of traffic signs in complex natural situations more effectively [10].

2. Literature review

With the development of advanced technologies such as the Internet of Things, artificial intelligence as well as cloud computing, the technology of intelligent driving will become more and more mature in the future. Traffic Sign Recognition (TSR), as the significant part of smart transportation systems, plays an irreplaceable role in safe travel. An effective TSR system can provide real and accurate real-time traffic information back to drivers in a timely manner, prompting them to take corresponding measures or to control the vehicle autonomously to achieve the purpose of avoiding danger, thus ensuring driving safety. Current research on key technology for traffic sign recognition is traffic sign detection technology. We have divided typical detection methods into four main categories based on the development history of traffic sign detection: colour-based methods, shape-based methods and deep learning-based methods.

2.1. Detection based on colour features

Traffic signs are generally classified into three categories: prohibition, indication and warning. These three types of signs all have distinct colour features. The RGB colour-based sign detection method started in the 1980s, when Akatsuka et al. [1] used a threshold segmentation algorithm on the RGB colour space to mark out the red, yellow and blue colours of traffic signs according to the threshold range of different colours to complete traffic sign detection. Zhangka et al. [14] counted all the colour information of the sign images and set the thresholds for red, yellow and blue based on the statistical information, i.e. using global colour features for sign image detection. Huang Zhiyong et al.[4] use the feature that RGB corresponds to a certain range of three-component difference values to obtain empirical thresholds after repeated experiments, and then segment the signs for detection, and this method does not require any multiplication operation.

2.2. Detection based on shape features

The most common method for shape detection is to use some form of the Hough transform. The Histogram of Oriented Gradients (HOG) is one of the acceptable choices for detecting shapes, with its representation of the direction of the gradient image. HOG essentially involves dividing the image into cells and accumulating a histogram of edge directions within those cells. Finally, features are generated to describe the object by combining the histogram entries.The HOG transform method has the advantage of rotational scaling invariance but is too computationally intensive. Paulo et al[10] first detect triangular and rectangular symbols by applying the Harris corner detector to a region of interest and then searching for the presence of corners in the six predefined control regions of that region. Gavrila uses the distance transform based template matching [8] for shape detection, firstly finding the edges in the original image, secondly creating a Distance Transform (DT) image and finally matching the template to the DT image.

2.3. Detection based on deep-learning

Deep learning extracts features through training and learning to achieve traffic sign detection, which is very

different from previous methods. The deep learning-based traffic sign detection method learns features by training big data, has strong feature expression capability, is not easily affected by external factors unrelated to traffic signs such as lighting and occlusion, and has stronger generalisation capability and higher accuracy than traditional traffic sign detection methods.

The target detection algorithm based on candidate region extraction has high detection accuracy. RCNN (Regions with CNN features) is a candidate region based target detection algorithm, which is advanced by Girshick et al.[11] It contains a rich feature hierarchy for accurate object detection and semantic segmentation by using a deep convolutional neural network (Convolutional Neural Network (CNN) to classify object proposals, achieving excellent object detection accuracy. However, because it repeatedly extracts and stores features for each candidate region, it costs a lot of computational time and storage resources. At the same time, RCNN uses region stretching to unify each candidate region into a size of 227 × 227, which affects the quality of the features extracted by CNN and reduces the detection accuracy. The FPN (Feature Pyramid Network) uses an RPN network on top of the feature pyramid to extract candidate regions, this method combines the depth feature and shallow feature to make prediction on multiple scales of the feature pyramid, which enhances the semantic of the shallow feature map and improves the detection accuracy of small targets.[12] propose the YOLO network, which frames detection as a single regression problem, predicting bounding box coordinates and class probabilities directly from the image.

3. Method

3.1. CNN feature extraction

Whether a classifier can achieve accurate classification is closely related to whether the features to be classified have target variability. Therefore, the feature description of the image to be recognized is the key to determine the classification effect, and feature extraction is also the core step of CNN[9].

3.1.1 Input layer

The input layer is the first layer of the entire neural network in a general convolutional neural network. Usually, all the images in the input layer must be of the same size in order to perform convolution and a series of operations on the image. The input layer can be either a colour image or a greyscale image. The length and width of the images in the input layer may vary, but in a classification task the image sizes are usually normalized to the same length and width.

3.1.2 Convolutional layer

A convolutional neural network model's fundamental layer is the convolutional layer and the convolution process is actually the process of feature extraction. In the convolutional layer, convolutional kernels of different sizes can be created with the response map of the previous layer, and these convolutional kernels are represented as some weight matrices[2]. Each convolution kernel function in the convolution process can be used as a filter to filter the input image, add a bias to the resulting response map and activate it with a non-linear activation function. The calculation of the convolution layer is as follows[2]:

$$X_j^l = f\left(\sum_{i \in M^j} X_i^{l-1} * K_{ij}^l + \mathrm{a}_j^l\right) \tag{1}$$

Here the first feature response map of the first layer is represented as the convolution kernel function, which is the bias, and the activation function.

3.1.3 Pooling layers

In most cases, the pooling layer is positioned behind the convolutional layer, and the pooling operation on the feature response map obtained from the convolutional layer can not only reduce the dimensionality, lessen the possibility of overfitting and simplify the complexity of network computing, thus saving computing time and improving the efficiency of training[9]; in addition, it can also maintain a certain invariance to the distortion of the input image such as translation, rotation and scaling when performing feature compression and extracting the main features This is a great advantage for image classification tasks.

3.1.4 Activation function

The activation function is the source of non-linearity in the network. Linear classifiers such as SVM can effectively solve binary classification problems, but for multi-classification problems, an important influencing factor on how to accurately classify is the non-linearity of the data. These are known as saturated non-linear

activation functions, which have a slow convergence rate and also suffer from gradient fading during the training process, causing the algorithm to fall into a local optimum in the process of finding the optimal parameters by gradient descent.

Rectified Linear Units (ReLU), in contrast to conventional non-linear activation functions, is an unsaturated non-linear activation function. In addition to its faster convergence rate, and secondly, its unilateral inhibition is more in line with the biological point of view, it can obtain sparse expressions, and its performance is better than that of traditional activation functions. Therefore, the algorithmic model in this paper chooses to use ReLU as the activation function to ensure that the network is non-linear in origin. In essence, the ReLU function is similar to a switch with a threshold of 0, which outputs 0 when the input value is less than 0, and itself when the input value is larger than 0.

$$R(x) = max(0, x) \tag{2}$$

3.2. CNN-based traffic sign recognition

Systems for recognizing traffic signs are a crucial component of intelligent driving systems and are essential for managing traffic sign maintenance and intelligent assisted driving. A convolutional neural network is a form of feed-forward neural network distinguished by the fact that each layer's neuron nodes only respond to neurons in their immediate surroundings of the previous layer (in a fully connected network, each neuron node responds to every node in the preceding layer). With different non-linear operations and pooling operations in between, a deep convolutional neural network model typically comprises of a variety of convolutional layers layered with a number of fully-connected layers. Back-propagation algorithms can also be used to train convolutional neural networks. Compared to other network models, the shared parameter nature of the convolutional operation allows the number of parameters to be optimised to be greatly reduced, improving the training efficiency and scalability of the model. Convolutional procedures have substantial advantages for the analysis and recognition of time series and picture data because they are primarily employed to handle data with grid-like structures.

This topic requires the use of CNN deep learning network to complete the recognition of traffic signs, the following mainly solve: data pre-processing, neural network design, algorithm operation debugging and other issues.

The overall design idea of the topic is shown in Figure 1.

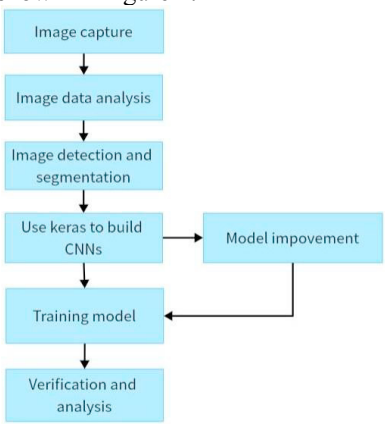

Figure 1 The overall design idea of the topic

3.2.1 Analysis of traffic sign data

Before modelling training it is first necessary to familiarise oneself with the data involved. The dataset used in this paper is from the web and consists of 43 different traffic signs commonly found on German roads. The traffic sign images are cropped from actual road images and are therefore under different lighting conditions, with a total of 50,000 images. The images are 32x32 pixels and in colour. The data set labels and the standard map of sign categories correspond to the following table:

A partial image of the training set is shown in Figure 2.

Figure 2 A partial image of the training set

A partial image of the test set is shown in Figure 3.

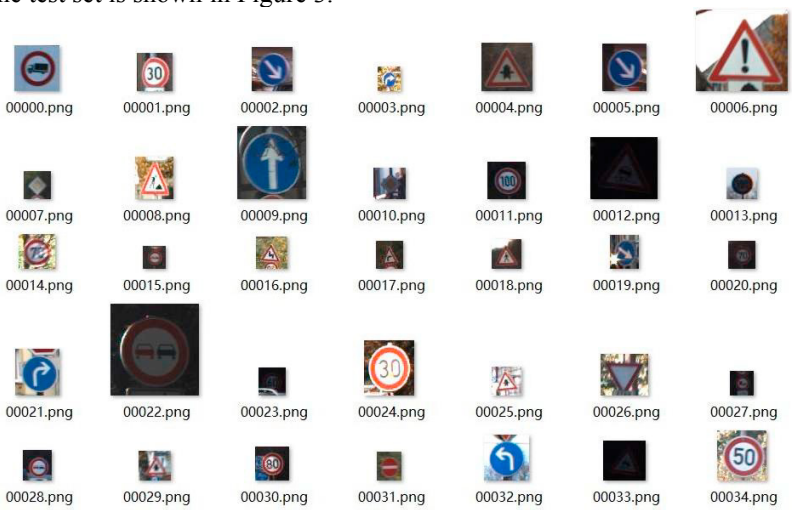

Figure 3 A partial image of the test set

3.2.2 HSV-based image processing and segmentation

Extracting the Region of Interest (ROI) is a necessary prerequisite for image recognition. Most of the current research is based on the 3 colours of red, yellow and blue of traffic signs for image segmentation [7]. Many colour spaces can be converted from the most basic and simple RGB colour space, such as HSV and HIS, but the RGB channel does not reflect the specific colour information of the object well, and for traffic sign images captured by in-vehicle cameras, inaccurate segmentation can easily occur because the R, G and B components are more correlated and tend to blend with the background objects when the illumination changes. In contrast to RGB space, HSV space is a highly intuitive representation of the lightness, darkness, hue and vibrancy of colours, allowing for easy comparison between colours. Therefore, in order to effectively avoid the effects of illumination changes and thus obtain obvious target regions, this paper uses a method of converting the acquired images into HSV colour space for colour segmentation, as expressed by Equation 11 [13].

$$\begin{cases} V = \frac{1}{\sqrt{3}} R + G + B \\ S = 1 - \frac{\sqrt{3}}{V} min(R, G, B) \\ H = \theta(B \leq G) or\ H = 360^{\circ} - \theta(B > G) \\ \theta = arccos \frac{(R-G)+(R-B)}{\sqrt[2]{(R-B)^2+(R-B)(R-G)}} \end{cases} \tag{3}$$

The HSV colour space was transformed through the image in RGB space, and the three main colour characteristics of traffic signs, namely blue, red, yellow, were first normalised to a threshold between the range [0, 1] for the three components of lightness V, saturation S, and hue H, according to their blue, red, and yellow colour characteristics. Then the optimal threshold range for different colour images is obtained through experimental verification, and finally the traffic signs with different colours are segmented, as shown in Table 1.

Table 1 Threshold range of hue, saturation and brightness of 3 colors of traffic signs

Color	Hue(H)	saturability(S)	Value(V)
Red	(0.026，0.039)	(0.18，1)	(0.12，1)
Yellow	(0.09，0.19)	(0.59，1)	(0.26，1)
Blue	(0.45，0.72)	(0.48，1)	(0.076，1)

In this paper, we have selected some images captured from life to train and test image processing and region segmentation. The following image is an example, Figure 4 is the original image without any processing.

Figure 4 original image

First, load the original image and convert it to an HSV channel (Figure 5). The image after extracting the blue area show in Figure 6.

Figure 5 HSV image after the original image is converted

Figure 6 The image after extracting the blue area

After image optimization processing, filtering processing, the image will be fuzzy, fuzzy next to the small white dot disappear (Figure 7).

Figure 8 Image after filtering

Then the processed image is binarized to form a black and white graph (Figure 8).

Figure 8 Image after binarization operation

Then the closed operation is used to connect the broken pixels(Figure 9).

Figure 9 Image after closing operation

The etching operation will erode the white pixels in the image to eliminate small specks (Figure 10).

Figure 10 Image after etching operation

Inflating causes the remaining white pixels to expand and grow back(Figure 11).

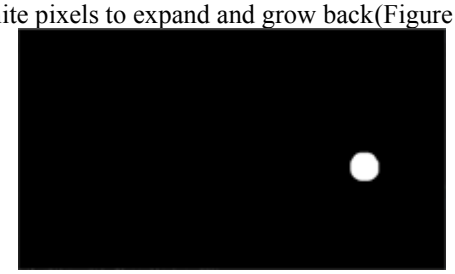

Figure 11 Image after expansion operation

In complex scenes, after optimisation, there will still be interference areas, which can be selectively cropped when looking for contours and cropping areas, for example, by first determining whether the area perimeter ratio of the area is the desired area to be cropped, I think it is easier and more efficient to filter by aspect ratio in traffic sign detection, because the basic shapes of common signs are square triangles, circles and squares. The area obtained after rectangular processing of the outline will be square (i.e. the outer rectangle of a triangle or circle is a square). Therefore, when cropping, those with aspect ratios close to 1 are kept, while those with too large a gap can be discarded.

In the dataset used for training and testing in this paper, although there is only one traffic sign image in each image of the dataset, there is still 10% of the background edge. Since the background edges are not useful to the classification algorithm and may even cause problems for subsequent recognition, the images are cropped in the first step of pre-processing to remove the redundant information in the background.

The results are shown in Figure 12. (a) and (c) are the images before cropping, and (b) and (d) are the images before cropping, which clearly show that the redundant background has been partially removed after cropping.

Figure 12 Comparison of before and after cropping of the dataset images

From the figure, it is clear that after the HSV-based modeling method, most of the irrelevant areas were removed through threshold segmentation, etc., to facilitate the next step of model training and testing.

It is clear from the figure that after processing by the HSV-based model approach, most of the irrelevant regions are removed by means of threshold segmentation, etc., to facilitate the next step of model training and testing.

3.2.3 CNN network architecture design

In this research, the CNN convolutional neural network was built using the Tensorflow + Keras architecture, which reduced the characteristics of the picture input while keeping the crucial and practical aspects to produce the prediction results. A high-level neural network API based on Tensorflow called Keras was created in Python. Because Keras is highly modular, minimalist, and extensible, it allows for easy and quick prototyping, support for CNN and RNN, or a combination of both, and effortless migration between CPU and GPU. Keras is designed to

support rapid experimentation. The main design flow chart is as show in Figure 13.

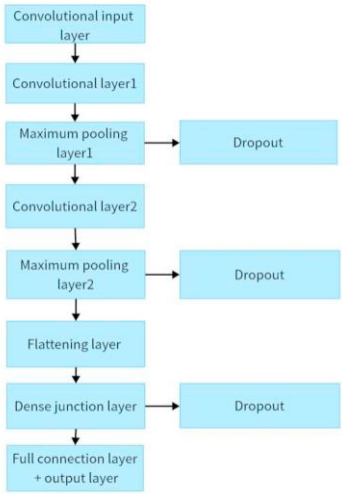

Figure 13 CNN neural network model design flow chart

For the existing CNN neural network, after the pooling layer, batch normalizing is introduced to process datasets. In the training of the network model, in general, with the increase of training time, the loss rate will gradually decrease until a stable range, but there are various possible factors to be taken into account to reduce the loss rate. In deep learning, model training is a very complex process. For example, if the attribute distribution of the input data has been altered only marginally in the early levels, it can be altered significantly in the later levels after the transfer.

Batch normalizing is a newly proposed method for dealing with differences in input data. This method normalizes the data in some local areas, so that the average value and variance of output results are 0 and 1, ensuring a uniform distribution of the output and to resolve the differences introduced by different images. Batch normalisation of the pooling layer follows the network model used in this paper. The improved CNN neural network design flowchart as show in Figure 14.

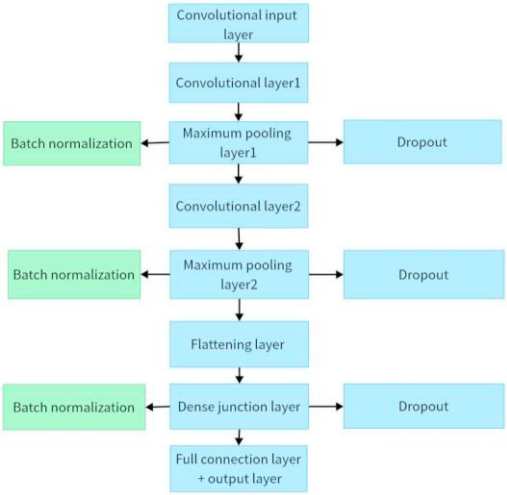

Figure 14 The improved CNN neural network design flowchart

4. Result

Initial CNN neural network model Train dataset, plots of training set (70%) accuracy vs. test set (30%) accuracy with number of iterations, and accuracy tested on the Test dataset.

Accuracy results (epochs=500), with the training set in blue and the Test set in orange.

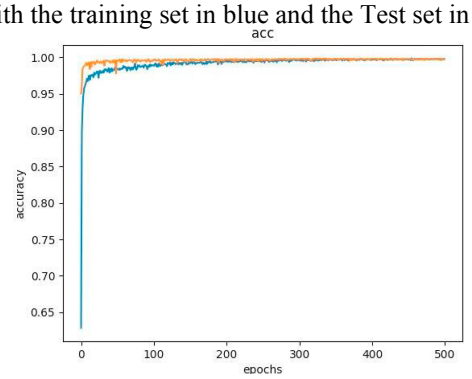

Figure 15 Train data set accuracy change map

Loss value results (epochs=500), blue is the training set, orange is the test set.

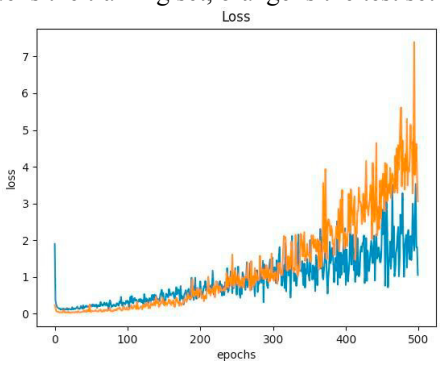

Figure 16 Train data set loss value change diagram

From Figure 15, it can be seen that the accuracy of both the training and test sets of the Train dataset increased with the number of iterations, and the accuracy of the test set increased faster than that of the training set at the beginning of the training period; from Figure 16, it can be seen that the loss decreased sharply at the beginning of the training period with the increase of the number of iterations, but it showed an increasing trend and fluctuated significantly with the increase of the number of iterations. The convergence of the model needs to be improved. The accuracy of the model was 97.56%, which is relatively accurate.

Added CNN neural network model Train dataset after Batch Normalization, plots of accuracy on the training set (70%) versus the test set (30%) with number of iterations, and accuracy tested on the Test dataset.

Accuracy results (epochs=500), with the Training set in blue and the Test set in orange.

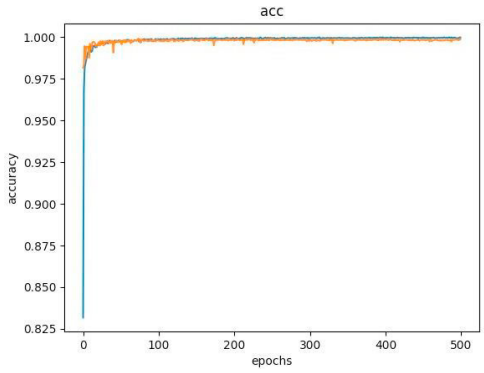

Figure 17 Train data set accuracy change map

Loss value results (epochs=500), with the Training set in blue and the Test set in orange.

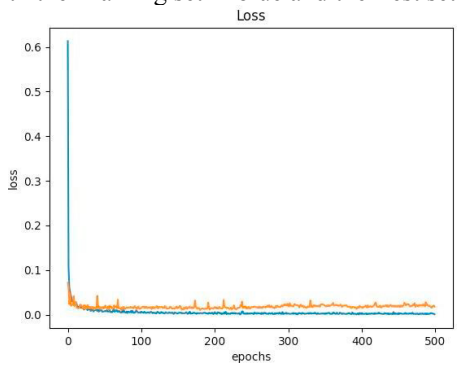

Figure 18 Change in loss values for the Train dataset

As can be seen from Figure 17, the accuracy of the improved model, both the training set and the test set of the Train dataset, keeps increasing with the number of iterations, and the trend tends to be consistent; as can be realized from Figure 18, the fluctuation of the improved loss value decreases and the loss gradually decreases, indicating that the improved model can converge better; at the same time, the training accuracy on the Test dataset has improved.

Dropout layer validation test results show in Table 2.

Table 2 Dropout validation

Group	Time/Iteration	Recognition accuracy
1	5s	0.9835312747426762
2	5s	0.9798891528107680
3	5s	0.9772763262074426
4	5s	0.9741092636579572

The comparison of the data in the second column of the above table, "Time per iteration", shows that there is little difference between the addition of the Dropout layer and the absence of the Dropout layer; the use of the Dropout layer in the test basically does not affect the recognition time; while comparing the recognition accuracy in the fourth column, it is obvious that the model uses the improved CNN structure with three Dropout layers can result in the highest recognition rate.

5. Discussion

This paper firstly presents a detailed introduction and analysis of the current research status and application areas of traffic sign detection and recognition technology, and then introduces the improvement of the traffic sign database, followed by the introduction of the application of CNN in image processing and traffic sign recognition, and the analysis of the current problems of traffic sign recognition.

In the algorithm implementation and improvement module, the framework and hardware and software environment used for the implementation of the designed network model are first introduced, followed by a detailed description of the code implementation of some important layers in the network model; and experiments are conducted to implement the basic CNN model.

After that, Batch Normalization was introduced to the existing CNN neural network to process the data set after the pooling layer, which effectively improved the problem of bouncing back after a sharp drop in the loss value of the neural network model and improved the convergence effect of the model; a comparison test was also conducted from the perspective of the number of Dropout layers, and the optimal number of layers, three, was obtained, proving that the model used The optimal number of Dropout layers is proved to be the optimal number.

In the end, the model identified in this paper has a high recognition accuracy (99.9%) and a short recognition time after training and testing, which proves that the improvement of the model in this paper is successful.

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